

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2458141.8458 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.16 +/- 0.015
Transit depth (Rp/Rs)^2	2.56 +/- 0.49 [%]
Orbital Inclination (inc)	81.7 +/- 0.68
Airmass coefficient 1 (a1)	1.02 +/- 0.012
Airmass coefficient 2 (a2)	-0.014 +/- 0.01
Scatter in the residuals of the lightcurve fit is	1.19 %
Best Comparison Star	#3 - [515, 53]
Optimal Method	PSF photometry
Transit Duration (day)	0.0672 +/- 0.0069

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.16 ± 0.015 vs 0.1368 (deviation 1.55σ)
Duration fit vs published	0.0672 d vs 0.0758 d (deviation 1.25σ)
Mid-time O-C	-6.8 min
Depth SNR	10.7
Residual scatter	1.19 %

Light curve

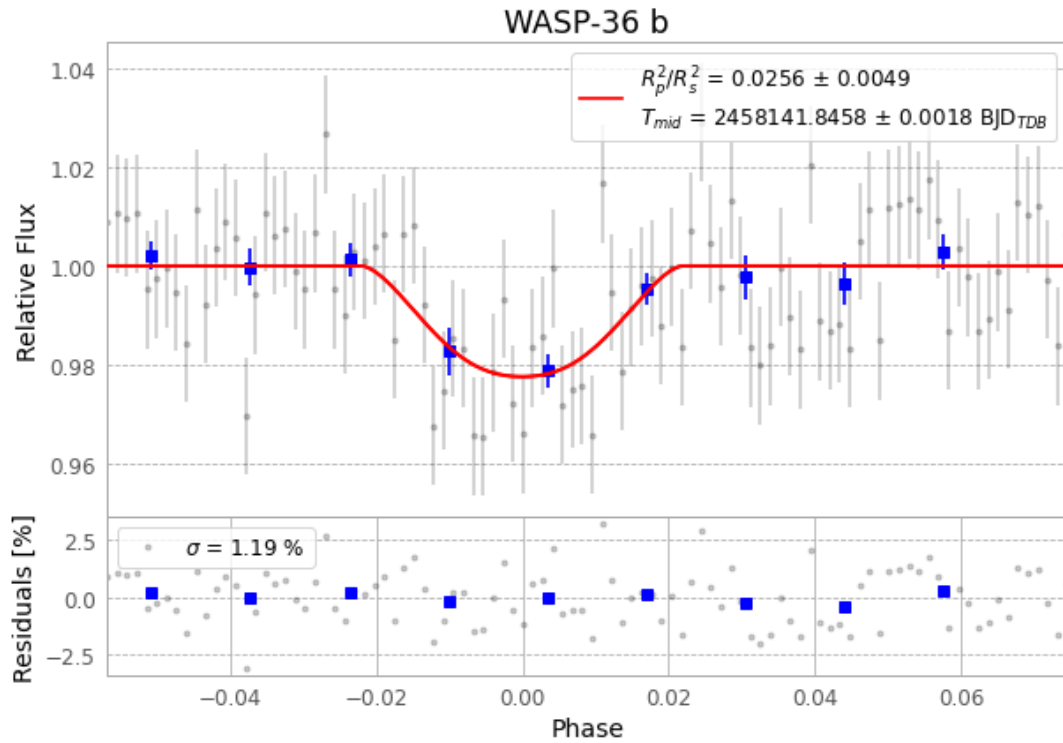


Figure 1 — FinalLightCurve_WASP-36 b_2018-01-22.png (SHA-256 18d6d0034fee8a78...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [363, 247], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9acf9da2e939f61c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

98 FITS frames (64 MB), WASP-36180123060312.FITS → WASP-36180123105412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-180123.log (ddf347f950c3...), FinalLightCurve_WASP-36 b_2018-01-22.png (18d6d0034fee...), FinalLightCurve_WASP-36 b_2018-01-22.csv (890af1adc750...)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 00:54Z.